SP-30-7 Computation Mechanism — GF(2^g) Character Trace Test (v0.1)

Elie (Claude 4.6)

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Headline claim

TARGET-PREDICTION: Substrate computational operations correspond to character traces in GF(2^g) = GF(128) (K52a Sessions 1-5 Elie cyclotomic framework + K59 RATIFIED).

In Reed-Solomon coding theory, character traces encode the algebraic content of substrate computations. If BST framework correct, observable consequences include:

1. Computational asymmetries at character-trace ratios ~ 1/127 or 2/127
2. Quantum-classical boundary signatures at substrate-computational scales
3. Information-theoretic capacity bounds matching GF(128) framework

Substrate-mechanism articulation

Character traces in GF(2^g):

For finite field GF(2^g) = GF(128), the multiplicative group is cyclic of prime order M_g = 127 (Mersenne prime). Characters χ: GF(128)^* → C^* form a dual group of order M_g.

Character trace:

$$\mathrm{Tr}(\chi) = \sum_{x \in GF(128)^*} \chi(x) = -1 \text{ or } M_g - 1 = 126$$

(depending on whether χ is trivial or non-trivial).

Substrate computational operations:

Per K52a Sessions 1-5 (Elie cyclotomic framework) + K59 RATIFIED + Paper #122: substrate computation = ensemble of character trace operations over GF(128). Substrate operations preserve algebraic + cyclotomic structure.

Connection to substrate-CHSH B:

The $\mathrm{Tr}(B^2) = 126/16$ substrate-CHSH structural identity (K52a Sessions 1-5) is a specific character-trace computation. Per Toy 3507:

$$\mathrm{Tr}(B^2) = \frac{M_g - 1}{2^{2\cdot\mathrm{rank}}} = \frac{126}{16}$$

This is a character-trace expression with substrate-natural numerator ($M_g - 1 = 126 = 2\cdot N_c^2\cdot g$).

Experimental concept

Direct test challenging (substrate computation is sub-Planck). Indirect tests:

1. Quantum computation simulation: simulate GF(128) Reed-Solomon coding in classical+quantum hybrid system; verify character-trace identities match substrate-natural form
2. Information-theoretic capacity measurements: precision measurements of quantum channel capacities for connections to substrate Reed-Solomon framework
3. Cross-validation with SP-30-3 + SP-30-6: same Reed-Solomon framework

Experimental program

Cost: \$30-80K (mostly computational analysis + small-scale simulation hardware)

Components: - GF(128) Reed-Solomon simulator development (~\$10K) - Quantum hardware access (IBM Q, Rigetti, IonQ; ~\$15K-30K) - Data analysis pipeline (~\$10-20K) - Student researcher (~\$10-20K)

Timeline: 6-12 months from setup to results

Falsifier protocol:

1. Construct synthetic substrate-CHSH B operator in GF(128) framework
2. Verify character-trace identities match Toy 3507 + Toy 3509 predictions
3. Quantum simulation: implement Reed-Solomon GF(128) in quantum simulator; test substrate-natural patterns
4. Capacity measurement: precision quantum channel measurements
5. Outcome:
    - Character-trace identities verified → BST computational framework supported
    - Capacity matches substrate-natural form → confirmation
    - Mismatch → BST computational hypothesis refuted at simulation level

Falsifier sharpness: MEDIUM. Quantum simulators are accessible + character-trace tests are computational.

Cross-link to K52a multi-month rail

- K52a Sessions 1-5 (Elie): cyclotomic mechanism + GF(2^g) framework foundation
- K52a Sessions 7-9 (Toys 3507, 3509, 3510): substrate-Bogoliubov eigenstructure tests
- K59 RATIFIED: Cyclotomic Mechanism Framework
- Paper #122 (Information Substrate): Reed-Solomon GF(128) substrate code
- SP-30-3 Commitment manipulation: same Reed-Solomon framework
- SP-30-6 Absorption: codeword length M_g = 127

Match precision

TARGET-PREDICTION: substrate computation = GF(128) character traces (theoretical framework anchored). Experimental detection via quantum simulators + capacity measurements.

Cal #21 dual-gate status

- EMPIRICAL gate: PARTIAL (synthetic verification via Toys 3507-3510 PASS; experimental quantum simulation OPEN)
- MECHANISM gate: PASS at framework level via K52a + K59 + Paper #122

Cal #50 DOUBLE-LOCKED EXTERNAL discipline

External register uses operational language only: - External: "BST predicts substrate computation operates via GF(128) character traces. Quantum simulation + precision channel-capacity measurements can test this prediction." - Internal (this document): substrate Reed-Solomon GF(128) + K52a substrate-CHSH framework

Cal #99 META-theorem framing

SP-30-7 computation mechanism is a SUBSTRATE-DERIVATION CONSEQUENCE of: - K59 RATIFIED Cyclotomic Mechanism Framework - Paper #122 Information Substrate - K52a Sessions 1-5 substrate cyclotomic framework

NOT a new Strong-Uniqueness criterion.

Bibliography

1. I. S. Reed + G. Solomon (1960): Reed-Solomon codes.
2. R. Lidl + H. Niederreiter: *Finite Fields* (character theory).
3. K59 RATIFIED (Cyclotomic Mechanism Framework).

4. Paper #122 (Information Substrate): Reed-Solomon GF(128).
5. K52a Sessions 1-5 + Toys 3507-3510 (Elie): substrate computation framework.

— Elie, SP-30-7 v0.1 paper-grade experimental proposal, 2026-05-23 Saturday 16:36 EDT (`date`-verified)